

Technical Report

A. Report Information

- **Môn học:**
- **Topic:** Author Collaborator Finder (ARXIV Co-Auth)
- **Lecturer:** Tran Hung Nghiep, nghiepth@uit.edu.vn
- **Group:** 2
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B. Report Content

Abstract

1. Introduction

1.1 Motivation

1.2 Problem Statement

This project addresses the problem of predicting future co-authorships in scientific collaboration networks. The goal is to identify which pairs of authors are most likely to collaborate next, based solely on historical co-authorship patterns extracted from the ArXiv hep-th (High Energy Physics – Theory) dataset.

Hassan & Hasan (2025) emphasize the importance of this research direction: “Link prediction in co-authorship networks is essential for understanding scientific collaboration patterns and identifying potential future collaborations.” [1]

However, despite extensive research in link prediction, several research gaps remain:

- Most link prediction studies focus on static co-authorship networks that ignore temporal evolution.
- Few works provide an interactive prototype system for real collaboration recommendation.
- Explainability is often missing, even though collaboration decisions require transparent justification.
- Temporal bias is common: many studies accidentally train on future data.

Wang et al. (2023) highlight this critical issue: “Predicting future collaborations requires explicitly incorporating temporal information and network evolution.” [2]

Therefore, our project aims to construct a temporal co-authorship dataset, implement and evaluate multiple link prediction models, and finally deploy an interactive and explainable collaborator recommendation system.

1.3 Objectives & Scope

Objectives

- Build a temporal co-authorship network from ArXiv metadata (1992–2003).
- Conduct structural network analysis (degree distribution, clustering, giant component).
- Implement and evaluate multiple link prediction heuristics: CN, AA, RA, PA, Community, and Hybrid models.
- Design a temporal train–test protocol to avoid data leakage.
- Develop a prototype application that recommends potential collaborators and explains the reasoning.

Scope

Included:

- Co-authorship network construction
- Temporal link prediction
- Heuristic methods
- Graph evaluation
- Prototype interface (Gradio)

Excluded:

- Supervised / ML-based link prediction (e.g., GNNs, embeddings)
- Large-scale author disambiguation
- Semantic features (keywords, topics), though recommended in prior work [5]
- Datasets outside hep-th

1.4 Contributions

- **Temporal Dataset Reconstruction:** We reconstruct a clean temporal co-authorship dataset from raw .abs metadata (1992–2003).
- **Network Analysis:** We analyze global structural properties, consistent with the findings that scientific networks are highly clustered and exhibit small-world characteristics.
- **Evaluation Framework with Strict Temporal Split**
- **Hybrid Link Prediction Model**
- **Explainable Recommendation System** (The prototype outputs):

- top-K suggested collaborators
- common neighbors
- shortest-path reasoning
- score by chosen method

2. Related Work

2.1 Background

2.2 Prior Work Comparison & Limitations

2.3 Positioning of This Work

3. Methodology

3.1 Overview of Approach

Our pipeline follows standard best practices for collaboration prediction:
 Data Extraction → Cleaning → Graph Construction → Temporal Split → Link
 Prediction → Evaluation → Prototype Interface

3.2 Data

The dataset consists of ArXiv hep-th abstracts from 1992–2003, producing thousands of papers and tens of thousands of collaboration edges.

Data Cleaning Steps:

- Extract author lines (Authors: / Author:)
- Remove affiliations, comments, escape characters
- Normalize separators (“and” → “,”)
- Form author pairs
- Assign earliest collaboration year
- Build edge list: author1, author2, year

Temporal Splitting:

- Training: edges $\leq T$ (e.g., 1998)
- Testing: edges $> T$
- Keep test edges whose nodes appear in train graph

3.3 System Design / Model Design

We implement several widely used unsupervised link prediction heuristics:

- **Common Neighbors (CN)**

- **Adamic–Adar (AA)**
- **Resource Allocation (RA)**
- **Preferential Attachment (PA)**
- **Community Similarity (COMM)**

We additionally introduce a **Hybrid AA + Community** score.

3.4 Training & Implementation Details

Tools and libraries used:

- Python
- NetworkX
- python-louvain
- Pandas
- Gradio
- Matplotlib/Seaborn

3.5 Evaluation Protocol

Metrics

- **AUC (Area Under ROC Curve)**
- **AP (Average Precision)**

Baselines

- CN, AA, RA, PA, Community

Proposed Method

- Hybrid (AA + Community)

4. Experiments and Results

4.1 Experimental Settings

- Dataset: ArXiv hep-th (1992–2003)
- Split year $T = 1998$
- Train graph: $\sim N_{\text{train}}$ nodes, $-- E_{\text{train}}$ edges
- Test set: $\sim E_{\text{test_pos}}$ positive edges
- Negative sampling: equal number $E_{\text{test_neg}}$
- Environment: Google Colab, Python 3.10

4.2 Quantitative Results

Method	AUC	AP
CN	0.575014	0.574686
AA	0.575013	0.574974
RA	0.575006	0.574848
PA	0.792814	0.798854
COMM	0.602028	0.597769
Hybrid	0.623575	0.621619

4.3 Qualitative Results

4.4 Empirical Analysis

5. Discussion

6. Conclusion and Future Work

6.1 Conclusion

6.2 Future Work

References

- [1] Hassan, D., & Hasan, M. A. (2025). Supervised Link Prediction in Co-Authorship Networks Based on Author Node-Based Features. *arXiv preprint arXiv:2505.21673*.
- [2] Wang, F., Dong, J., Lu, W., & Xu, S. (2023). Collaboration prediction based on multilayer all-author tripartite citation networks: A case study of gene editing. *Journal of Informetrics*, 17(1), 101374.
- [3] Chuan, P. M., Son, L. H., Ali, M., Khang, T. D., Huong, L. T., & Dey, N. (2018). Link prediction in co-authorship networks based on hybrid content similarity metric. *Applied Intelligence*, 48(8), 2470-2486.
- [4] Cho, H., & Yu, Y. (2018). Link prediction for interdisciplinary collaboration via co-authorship network. *Social Network Analysis and Mining*, 8(1), 25.
- [5] Yuliansyah, H., Othman, Z. A., & Bakar, A. A. (2025). Co-authorship prediction method based on degree of gravity and article keywords similarity. *Physica A: Statistical Mechanics and its Applications*, 665, 130511.
- [6] Chennupati, N. (2021). Recommending collaborations using link prediction.
- [7] *Link Property Prediction* (OGB collaboration dataset)

Appendices

A. Technical Details

B. Project Planning

7.1 Timeline

- Weeks 1–2: Literature review, dataset exploration
- Weeks 3–4: Metadata parsing, graph construction
- Weeks 5–6: Implementation of heuristics, community detection
- Week 7: Temporal split & evaluation
- Week 8: Hybrid model + UI prototype
- Week 9: Experiments, results & analysis
- Week 10: Report writing & presentation preparation

7.2 Team Responsibilities

- Ha Nhat Thai: Data extraction & cleaning
Graph construction & heuristics implementation
- Nguyen Khanh Tuan: Evaluation & hybrid model
UI prototype & report

7.3 Current Progress

- Data cleaning: Partially completed.

Several author-name records are still noisy (e.g., affiliation residues, escaped characters, concatenated comments). Further refinement of the cleaning pipeline is required to improve data consistency.

- Graph building: Completed.

The co-authorship graph for 1992–2003 has been successfully constructed, but may require regeneration once the improved cleaned dataset is available.

- Baseline heuristics: Implemented & evaluated.

Current AUC/AP scores are acceptable but not yet optimal due to remaining data noise and incomplete author normalization.

- Hybrid model: Implemented.

The hybrid (AA + Community) model is still capped by the quality of the underlying data.

- UI prototype: Deployed via Gradio.

The interface functions correctly and provides suggestions with explanations, but future updates may improve usability and scalability.

- Report & slides: In drafting.
- Future tasks:

Improve the robustness of data cleaning (eliminate formatting noise, parenthetical affiliations, “Comments” leakage, and escape strings).

Regenerate the temporal dataset and recompute all metrics.

Re-evaluate all baseline and hybrid models on the newly cleaned dataset to achieve higher AUC/AP scores.

Finalize the report, polish the slides, and record the demo video.